

Harshwardhan Pandey

+91 73508 54754 | harshpandey145@gmail.com | GitHub: [Harsh-debug04](#) |

Objective

Aspiring data science and machine learning professional passionate about solving real-world problems using AI, data analytics, and emerging technologies.

Education

B.Tech in Computer Science Engineering

Manipal University Jaipur, Rajasthan

2022 – Expected Graduation: 2026

Technical Skills

- **Languages:** Python, C++
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Gradient Boosting (GBM), Model Optimization, Explainable AI
- **NLP & Generative AI:** BERT, LLM Models, LangChain, Retrieval-Augmented Generation (RAG), OpenCV
- **Data Analytics & Visualization:** Power BI, Tableau, Excel, Canva
- **Databases:** PostgreSQL, MySQL, MongoDB, Prisma ORM
- **DevOps & Tools:** Git, Jupyter Notebook, Google Colab, Kaggle, Azure
- **Concepts:** Data Structures & Algorithms (DSA), OOP, NLP, Data Visualization

Work Experience

- **Full Stack & AI Developer – AgriCart Farmtech Private Limited** July 2025 – Present
 - Built and maintained Magento-based frontend for e-commerce startup (B2B2C model)
 - Fine-tuned LLMs on agriculture datasets to build an AI-powered chatbot for Indian farmers & enterprises
 - Collected, cleaned, and structured domain-specific agricultural data for model training
- **Blockchain Developer – Deloitte (Capstone Program)** April – July 2025
 - Designed blockchain-powered audit trail system for supply chain transparency.
 - Implemented smart contracts and immutable ledger systems to improve traceability & trust.
- **Research Intern – IIT Roorkee** June 2023
 - Contributed to sustainable development research under IIT Roorkee’s SPARK Internship Program.
 - Conducted environmental data analysis and impact studies.

Projects

- **Predicting Power Outages in Delhi Using Weather and Substation Data**
Forecasted potential power failures using historical weather data and network topology analysis to assist grid optimization.
- **Interpretable Disease Diagnosis**
Used explainable AI techniques to interpret black-box ML models in healthcare and provide transparent diagnosis support.
- **Population Density Estimation Using Satellite Imagery**
Designed a simple model using image classification techniques to estimate population density from satellite images.
- **Sentiment Analysis Using NLP**
Created a classifier using BERT to extract and categorize sentiments from user reviews and social media text.
- **Vision Transformers vs. CNN**
Conducted a comparative study of Vision Transformers and Convolutional Neural Networks for image classification.

Research Experience

- **Visualizing Pneumonia Detection in Chest X-Ray Images Using Explainable AI**
Developed visualization methods to interpret AI predictions for pneumonia detection in radiographic images.
- **GAN-LSTM in Watermarking**
Explored hybrid neural architectures to improve robustness and invisibility in digital watermarking techniques.
- **Deep Reinforcement Learning-Based UAV Navigation**
Reviewed current techniques for autonomous UAV navigation and conflict resolution in unknown environments.

Leadership & Achievements

- **Smart India Hackathon – Finalist**
- **IEEE Hackathon – Finalist**
- **Capstone Program – Finalist**